

Diagnostic report: Mg I

Computed levels: 192
Matched to NIST: 39
Computed E1 lines: 9999
Level-fit RMS ($E_{\text{calc}} - E_{\text{obs}}$): 85.8 cm⁻¹

Energy-level fit vs Bob Kurucz ($E_{\text{obs}} \leq 60000$ cm⁻¹, offset-removed):

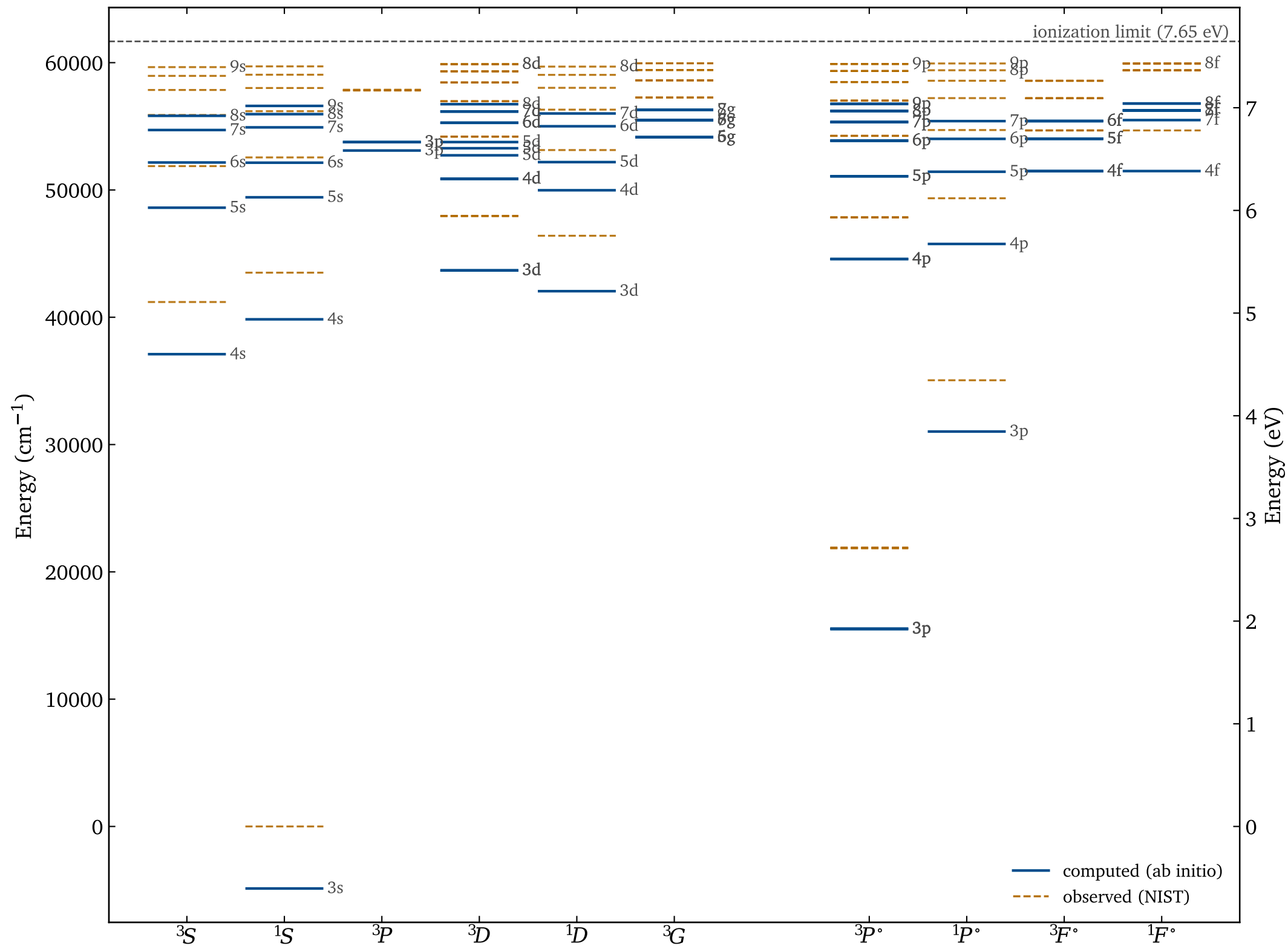
	N	median E-0	RMS	90th	max
this fit	102	3.2	86	126	520
Bob (RCE)	118	0.0	18	22	78

bulk (median) is near-Bob; the ~3x RMS gap is the singlet
3s.nd 1D / near-IE tail (unobserved-perturber wall), not
data starvation -- the basis matches Bob's b1200{e,o} deck.

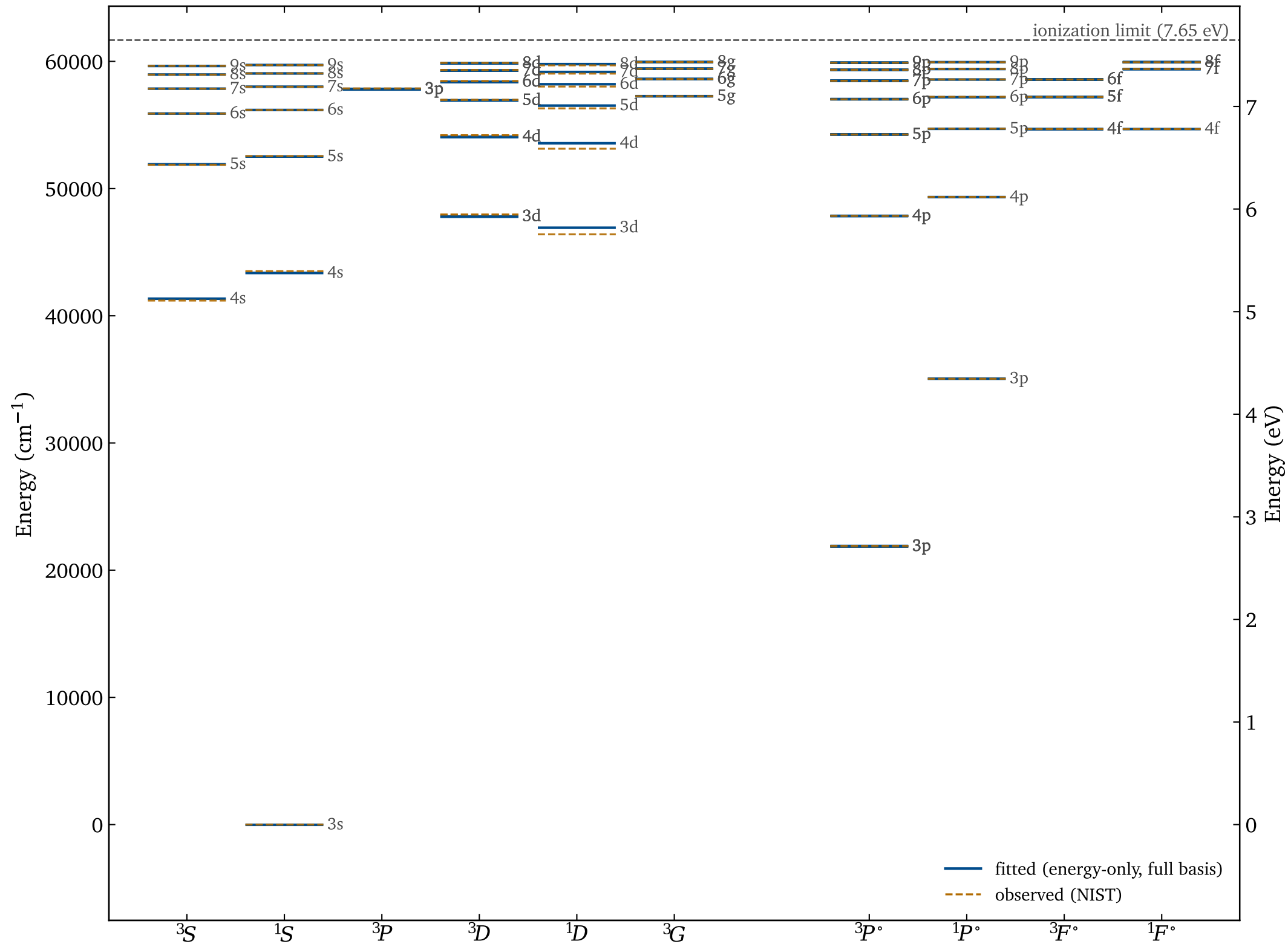
Strongest computed lines:

lambda(A)	log gf	upper term
123623.32	+1.406	(2S) 3H
190436.94	+1.360	(2S) 3H
123623.32	+1.338	(2S) 1H
123622.63	+1.337	(2S) 3H
277772.61	+1.329	(2S) 3H
190436.93	+1.292	(2S) 1H
190436.11	+1.291	(2S) 3H
277772.59	+1.262	(2S) 1H

Mg I: term diagram - ab initio vs observed



Mg I: term diagram - fitted vs observed



Mg I: levels — ab initio & fitted vs observed (cm⁻¹)

config	term	J	E_{obs}	E_{abinit}	Δ_{abinit}	E_{fit}	Δ_{fit}
3s2	¹ S	0	0.0	-4873.0	-4873.0	-21.1	-21.1
3s.3p	³ P°	0	21850.4	15503.0	-6347.4	21858.9	+8.5
3s.3p	³ P°	1	21870.5	15517.0	-6353.5	21872.9	+2.4
3s.3p	³ P°	2	21911.2	15544.0	-6367.2	21900.9	-10.3
3s.3p	¹ P°	1	35051.3	31022.0	-4029.3	35051.9	+0.6
3s.4s	³ S	1	41197.4	37100.0	-4097.4	41343.9	+146.5
3s.4s	¹ S	0	43503.3	39838.0	-3665.3	43365.9	-137.5
3s.3d	¹ D	2	46403.1	42049.0	-4354.1	46922.9	+519.8
3s.4p	³ P°	0	47841.1	44570.0	-3271.1	47846.9	+5.8
3s.4p	³ P°	1	47844.4	44572.0	-3272.4	47849.9	+5.5
3s.4p	³ P°	2	47851.2	44578.0	-3273.2	47855.9	+4.7
3s.3d	³ D	2	47957.0	43677.0	-4280.0	47800.9	-156.2
3s.3d	³ D	3	47957.0	43678.0	-4279.0	47800.9	-156.2
3s.3d	³ D	1	47957.1	43677.0	-4280.1	47800.9	-156.2
3s.4p	¹ P°	1	49346.7	45759.0	-3587.7	49330.9	-15.9
3s.5s	³ S	1	51872.5	48606.0	-3266.5	51908.9	+36.3
3s.5s	¹ S	0	52556.2	49424.0	-3132.2	52518.9	-37.3
3s.4d	¹ D	2	53134.6	49977.0	-3157.6	53563.9	+429.2
3s.4d	³ D	3	54192.3	50871.0	-3321.3	54065.9	-126.4
3s.4d	³ D	2	54192.3	50871.0	-3321.3	54065.9	-126.4
3s.4d	³ D	1	54192.3	50871.0	-3321.3	54065.9	-126.5
3s.5p	³ P°	0	54248.8	51078.0	-3170.8	54251.9	+3.1
3s.5p	³ P°	1	54250.1	51079.0	-3171.1	54252.9	+2.8
3s.5p	³ P°	2	54252.7	51081.0	-3171.7	54254.9	+2.1
3s.4f	³ F°	2	54676.7	51477.0	-3199.7	54675.9	-0.8
3s.4f	³ F°	3	54676.7	51477.0	-3199.7	54675.9	-0.8
3s.4f	³ F°	4	54676.8	51477.0	-3199.8	54675.9	-0.9
3s.4f	¹ F°	3	54676.4	51484.0	-3192.4	54679.9	+3.4
3s.5p	¹ P°	1	54706.5	51431.0	-3275.5	54698.9	-7.7
3s.6s	³ S	1	55891.8	52158.0	-3733.8	55899.9	+8.1
3s.6s	¹ S	0	56186.9	52142.0	-4044.9	56179.9	-7.0
3s.5d	¹ D	2	56308.4	52189.0	-4119.4	56523.9	+215.5
3s.5d	³ D	3	56968.2	53763.0	-3205.2	56933.9	-34.3

Mg I: levels — ab initio & fitted vs observed (cm⁻¹)

config	term	J	E_{obs}	E_{abinit}	Δ_{abinit}	E_{fit}	Δ_{fit}
3s.5d	3D	1	56968.3	52722.0	-4246.3	56933.9	-34.4
3s.6p	$^3P^\circ$	0	57017.1	53868.0	-3149.1	57027.9	+10.8
3s.6p	$^3P^\circ$	1	57017.7	53868.0	-3149.7	57028.9	+11.1
3s.6p	$^3P^\circ$	2	57019.0	53869.0	-3150.0	57029.9	+10.8
3s.6p	$^1P^\circ$	1	57215.0	54014.0	-3201.0	57182.9	-32.1
3s.5f	$^3F^\circ$	2	57204.2	54024.0	-3180.2	57202.9	-1.4
3s.5f	$^3F^\circ$	3	57204.3	54024.0	-3180.3	57203.9	-0.4
3s.5f	$^3F^\circ$	4	57204.3	54024.0	-3180.3	57203.9	-0.4
3s.5f	$^3F^\circ$	3	57204.2	54029.0	-3175.2	57206.9	+2.7
3s.5g	3G	4	57262.8	54141.0	-3121.8	57262.9	+0.1
3s.5g	3G	3	57262.8	54141.0	-3121.8	57262.9	+0.1
3p2	3P	0	57812.8	53098.0	-4714.8	57793.9	-18.9
3p2	3P	1	57833.4	53763.0	-4070.4	57808.9	-24.5
3p2	3P	2	57873.9	53763.0	-4110.9	57839.9	-34.1
3s.7s	3S	1	57855.2	54709.0	-3146.2	57855.9	+0.7
3s.7s	1S	0	58009.4	54918.0	-3091.4	58009.9	+0.5
3s.6d	1D	2	58023.2	55004.0	-3019.2	58205.9	+182.6
3s.6d	3D	3	58442.8	55275.0	-3167.8	58385.9	-57.0
3s.6d	3D	2	58442.9	55275.0	-3167.9	58385.9	-57.0
3s.6d	3D	1	58442.9	55275.0	-3167.9	58385.9	-57.0
3s.7p	$^3P^\circ$	0	58476.7	55335.0	-3141.7	58479.9	+3.2
3s.7p	$^3P^\circ$	1	58477.0	55336.0	-3141.0	58479.9	+2.9
3s.7p	$^3P^\circ$	2	58477.8	55336.0	-3141.8	58479.9	+2.1
3s.6f	$^3F^\circ$	2	58575.5	55413.0	-3162.5	58574.9	-0.7
3s.6f	$^3F^\circ$	3	58575.5	55413.0	-3162.5	58574.9	-0.7
3s.6f	$^3F^\circ$	4	58575.5	55413.0	-3162.5	58574.9	-0.7
3s.6f	$^3F^\circ$	3	58575.5	55416.0	-3159.5	58576.9	+1.4
3s.7p	$^1P^\circ$	1	58580.2	55409.0	-3171.2	58572.9	-7.4
3s.6g	3G	4	58610.8	54141.0	-4469.8	58610.9	+0.1
3s.6g	3G	3	58610.8	55483.0	-3127.8	58610.9	+0.1
3s.6g	3G	5	58610.8	54141.0	-4469.8	58610.9	+0.1
3s.8s	3S	1	58962.7	55824.0	-3138.7	58963.9	+1.1
3s.8s	1S	0	59053.5	55955.0	-3098.5	59051.9	-1.6

Mg I: levels — ab initio & fitted vs observed (cm⁻¹)

config	term	J	E_{obs}	E_{abinit}	Δ_{abinit}	E_{fit}	Δ_{fit}
3s.7d	3D	3	59318.8	56167.0	-3151.8	59277.9	-40.9
3s.7d	3D	2	59318.8	56167.0	-3151.8	59277.9	-40.9
3s.7d	3D	1	59318.8	56167.0	-3151.8	59277.9	-40.9
3s.8p	$^3P^\circ$	0	59342.5	56204.0	-3138.5	59343.9	+1.4
3s.8p	$^3P^\circ$	1	59342.5	56204.0	-3138.5	59343.9	+1.4
3s.8p	$^3P^\circ$	2	59342.5	56205.0	-3137.5	59343.9	+1.4
3s.7f	$^1F^\circ$	2	59400.8	56249.0	-3151.8	59400.9	+0.1
3s.7f	$^1F^\circ$	3	59400.8	56249.0	-3151.8	59400.9	+0.1
3s.7f	$^1F^\circ$	4	59400.8	55488.0	-3912.8	59400.9	+0.1
3s.7f	$^1F^\circ$	3	59400.8	56251.0	-3149.8	59401.9	+1.1
3s.8p	$^1P^\circ$	1	59403.2	--	--	59399.9	-3.3
3s.7g	3G	4	59423.5	55483.0	-3940.5	59423.9	+0.3
3s.7g	3G	3	59423.5	56293.0	-3130.5	59423.9	+0.3
3s.7g	3G	5	59423.5	55483.0	-3940.5	59423.9	+0.3
3s.9s	3S	1	59649.2	--	--	59639.9	-9.3
3s.9s	1S	0	59707.1	56603.0	-3104.1	59716.9	+9.8
3s.8d	1D	2	59690.0	--	--	59785.9	+95.9
3s.8d	3D	3	59881.2	56738.0	-3143.2	59849.9	-31.3
3s.8d	3D	2	59881.2	--	--	59849.9	-31.3
3s.8d	3D	1	59881.2	--	--	59849.9	-31.3
3s.9p	$^3P^\circ$	0	59897.9	56761.0	-3136.9	59896.9	-1.0
3s.9p	$^3P^\circ$	1	59897.9	--	--	59896.9	-1.0
3s.9p	$^3P^\circ$	2	59897.9	56762.0	-3135.9	59896.9	-1.0
3s.8f	$^1F^\circ$	2	59935.4	--	--	59934.9	-0.5
3s.8f	$^1F^\circ$	3	59935.4	56790.0	-3145.4	59934.9	-0.5
3s.8f	$^1F^\circ$	4	59935.4	56249.0	-3686.4	59934.9	-0.5
3s.8f	$^1F^\circ$	3	59935.4	56791.0	-3144.4	59935.9	+0.5
3s.9p	$^1P^\circ$	1	59936.6	--	--	59940.9	+4.2
3s.8g	3G	4	59950.9	55483.0	-4467.9	59950.9	+0.0
3s.8g	3G	5	59950.9	56293.0	-3657.9	59950.9	+0.0

